

Air Quality Intelligence System

Advanced Database Management Systems

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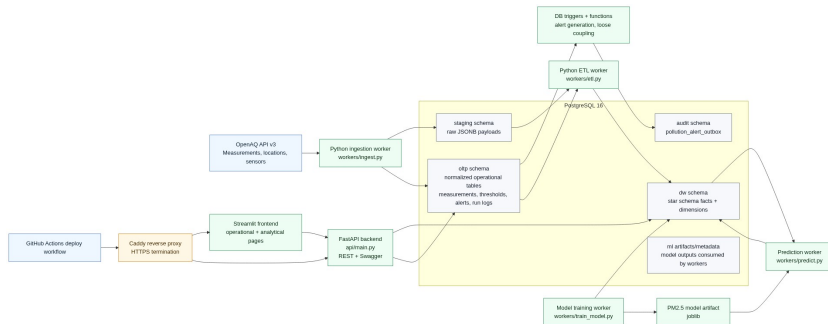
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Problem Statement

- Urban air pollution is a growing health concern
- Publicly available sensor data (OpenAQ) lacks real-time alerting
- Need: automated ingestion, threshold-based alerting, and operational oversight

Goal: Build a production-ready DBMS-backed monitoring system demonstrating: triggers, transactions, partitioning, indexing, backups, and an operational GUI.

Architecture Overview



Technologies Used

- PostgreSQL 16
- PL/pgSQL triggers
- Range partitioning
- Explicit transactions
- CHECK / FK constraints
- Python 3.12
- FastAPI (REST)
- Streamlit (GUI)
- Docker Compose
- Caddy (HTTPS)
- GitHub Actions (CI/CD)

Demo: Operational GUI — Stations & Measurements

- Station Explorer page: select city → station → parameter
- Shows time-series chart and raw measurement table
- Data comes from `oltp.measurement_raw` (partitioned)

Live demo: open <https://mw79on-demo.online>, navigate to Station Explorer

Demo: Threshold Management

- CRUD interface for `oltp.threshold_rule`
- Each rule: city + pollutant + severity level + minimum value
- Unique constraint: `(parameter_id, city, warning_level)`

Live demo: add/edit a threshold rule, show it persisted in the table.

Demo: Trigger-Based Alert Generation

- Trigger: `trg_measurement_generate_pollution_alert`
- Fires AFTER INSERT on `measurement_raw`
- Matches value against active threshold rules → inserts alert row
- Second trigger enqueues event into `audit.pollution_alert_outbox`

Live demo: insert measurement → observe new alert appearing.

Demo: Alert Review & Transaction Safety

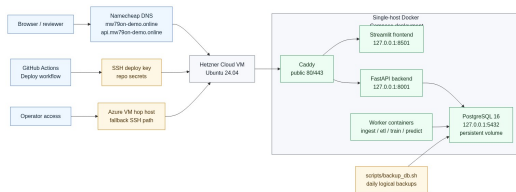
- Alerts page: filter by status / severity
- Transition: open → reviewed → closed (with notes)
- Ingestion uses explicit transactions with savepoints
- Partial batch failure → atomic rollback, no corrupt data

Live demo: review an alert; show ingestion log for transaction evidence.

Demo: Indexing, Partitioning, Backup

- `measurement_raw`: range-partitioned by month (25+ partitions)
- B-tree indexes on FKs and hot query columns
- `scripts/backup.sh`: `pg_dump` with rotation & optional off-site
- System Status page shows ingestion run history

- Hetzner Cloud VM (Ubuntu 24.04)
- Docker Compose: 5 services
- Caddy: automatic HTTPS (Let's Encrypt)
- GitHub Actions: push → deploy
- Cron scheduling:
 - Every 3h: ingest + ETL
 - Daily: predictions
 - Daily: backup with rotation



Summary & Lessons Learned

- PostgreSQL triggers provide powerful event-driven logic
- Partitioning enables efficient time-range queries and archival
- Explicit transactions with savepoints ensure data integrity
- Docker Compose + Caddy = simple production deployment
- CI/CD keeps the system always up-to-date

Live system: <https://mw79on-demo.online>

Source code: [https://github.com/dobosipeter/
advanced-data-warehouse-and-database-management-systems-submission](https://github.com/dobosipeter/advanced-data-warehouse-and-database-management-systems-submission)